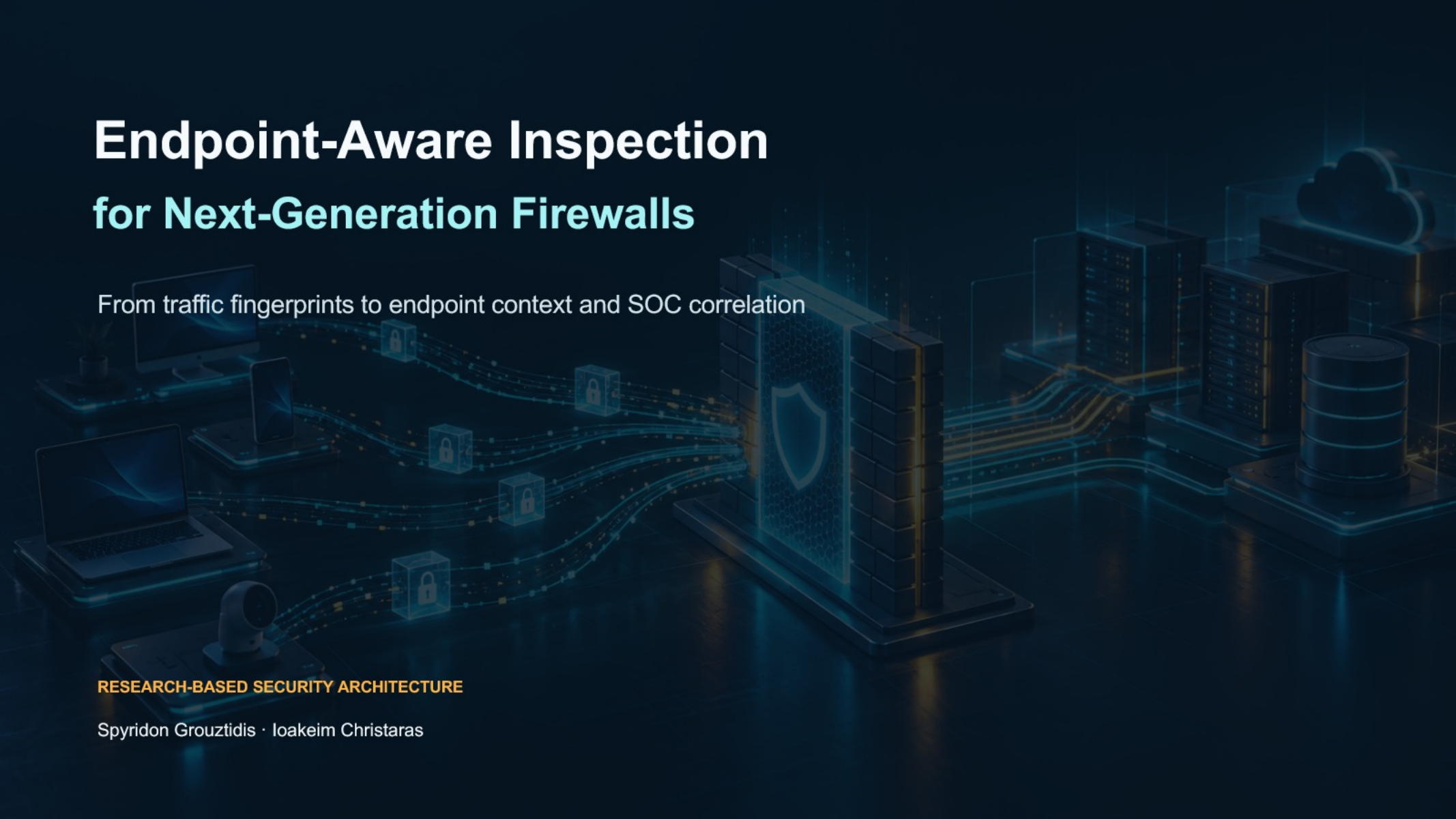
Endpoint-Aware Inspection

for Next-Generation Firewalls

From traffic fingerprints to endpoint context and SOC correlation



RESEARCH-BASED SECURITY ARCHITECTURE

Spyridon Grouztidis · Ioakeim Christaras

The firewall sees traffic, but the risk depends on the endpoint

One network pattern can be harmless for one application and dangerous for another.



Missing context

Process identity · software version · device posture · asset state

Endpoint awareness narrows policy decisions and improves alert context. It does not replace traffic inspection.

Evidence base and analytical boundary

Research taxonomy

Heino et al. divide endpoint application identification into passive and active methods. Their survey reviews academic work and public implementation evidence.

Zero Trust context

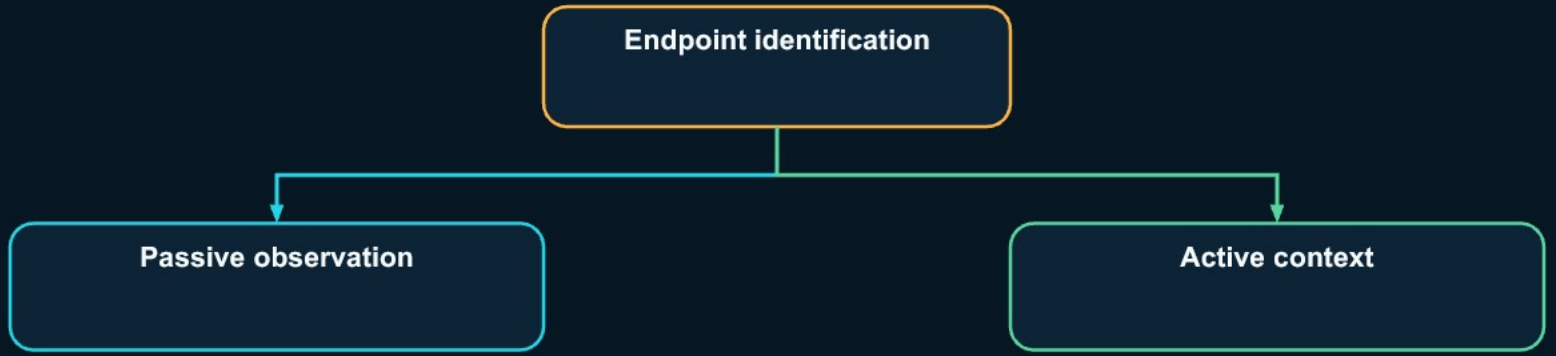
NIST SP 800-207 treats policy enforcement as a decision informed by identity, asset state, diagnostics, and policy. Endpoint context can become one input to that decision.

Portfolio interpretation

The architecture and SOC rule in this repository are design artifacts. They have not been deployed, benchmarked, or validated against a production firewall.

Strong evidence labels make the design useful without overstating implementation maturity.

Endpoint identification taxonomy



Broad coverage, weaker identity

Stronger context, narrower reach

Passive methods provide scale, but identity remains probabilistic

Protocol metadata

HTTP User-Agent and visible protocol fields are inexpensive signals, but clients can spoof them and encryption reduces visibility.

TLS fingerprints

ClientHello and ServerHello features can cluster clients or servers. Shared TLS libraries and evasion create collisions and false attribution.

Flow features

Packet sizes, timing, and bursts can support classification without payload inspection. Some methods require complete flows and model maintenance.



A fingerprint supports an inference. It should never become the sole proof of application identity.

Operational use: clustering, hunting, enrichment, and confidence scoring

Active methods improve context at the cost of reach and maintenance

Active scanning

Best suited to exposed server services. Results age between scans, probing requires authorization, and comprehensive discovery consumes time and resources.

Examples: service detection, version discovery, authorized vulnerability scanning.

CMDB and inventory

Adds ownership, expected software, business criticality, and management state. Its value depends on coverage, synchronization, and governance.

Main failure mode: stale or incomplete records.

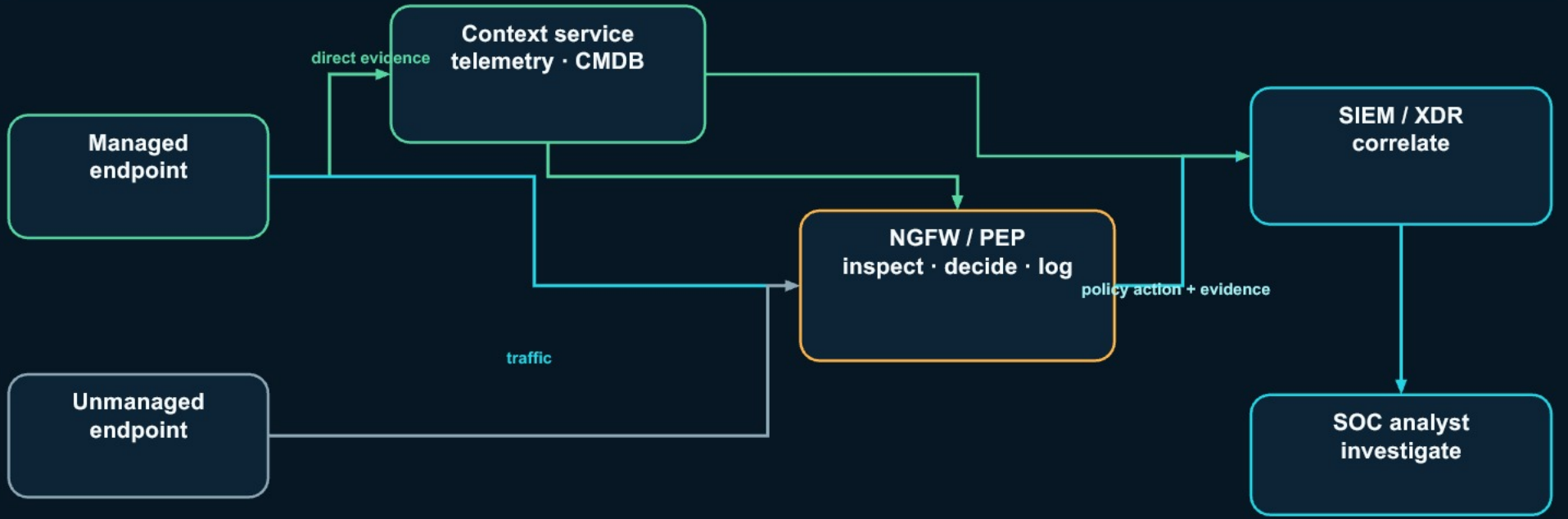
Endpoint telemetry

Maps a connection to a local process and device. It offers strong attribution on managed endpoints but requires a trusted sensor, secure transport, and privacy controls.

Main constraint: unmanaged endpoints remain outside coverage.

Freshness and provenance must travel with every context assertion.

Reference architecture: combine network evidence with endpoint context



Operational comparison

Method	Coverage	Freshness	Identity confidence	Main constraint
HTTP metadata	Low	Live	Low	Spoofing and encryption
TLS fingerprint	High	Live	Low–medium	Collisions and protocol change
Flow classification	High	Delayed	Low–medium	Full-flow and model cost
Active scanning	Servers	Periodic	Medium	Authorization and staleness
CMDB	Managed assets	Variable	Medium	Record quality
Endpoint telemetry	**Managed endpoints**	**Near real time**	**High**	**Agent trust and privacy**

The best operational design combines coverage with stronger evidence and records confidence explicitly.

Detection use case: suspicious TLS fingerprint with process context



Severity logic

Higher risk
Unsigned process
Unmanaged device
Critical asset

Lower risk
Approved signer
Common process
Expected destination

The fingerprint starts the investigation. Endpoint evidence determines whether the signal deserves priority.

Operational risks and required controls

Evasion and misattribution

Fingerprints can collide or be imitated. Preserve raw evidence, use prevalence baselines, and avoid identity claims from a single network feature.

Stale context

CMDB records and scan results age. Timestamp assertions, set expiration thresholds, and define fallback policy when context is missing.

Telemetry trust

A compromised endpoint can falsify local metadata. Authenticate sensors, protect transport, and compare endpoint claims with independent network observations.

Privacy and governance

Process, user, and connection metadata can be sensitive. Minimize fields, restrict access, document purpose, and enforce retention limits.

Enforcement requires multiple signals, measurable confidence, and a reversible rollout.

Implementation roadmap

01

Baseline visibility

Inventory log sources, normalize connection fields, and measure join coverage.

02

Controlled pilot

Enable endpoint-to-network correlation for a managed device group in alert-only mode.

03

Tune confidence

Baseline fingerprints and processes, test benign edge cases, and document exceptions.

04

Selective enforcement

Apply policy only where evidence quality, rollback, and ownership are clear.

Success measures Join coverage · context freshness · false-positive rate · analyst time · rollback success

Start with observability. Move to blocking only after evidence quality is measurable.

Conclusion

Endpoint awareness improves NGFW decisions when evidence from the network and the endpoint is combined.

Passive visibility

Broad coverage and immediate observation, with probabilistic attribution.

Active context

Stronger attribution and asset state, with deployment and freshness constraints.

SOC correlation

A practical layer that preserves evidence, confidence, and analyst review.

No silver bullet: use layered signals, explicit confidence, and controlled enforcement.